

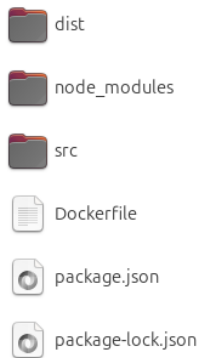
PROGRAM 2: Develop a multistage docker file for container orchestration

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SECTION: A

Directory structure:



Step 1: Create a Dockerfile with the following

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$ cat Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json package-lock.json ./
RUN npm install

COPY . .
RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000

CMD ["node", "dist/index.js"]
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$
```

Step 2: Create [index.js](#) inside the src directory

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/Dev0ps/p2$ cat src/index.js
const express = require('express');
const app = express();

app.get('/', (req, res) => {
    res.send("Hello from Node.js application.");
});

app.listen(3000, () => {
    console.log("Listening at port 3000");
});
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/Dev0ps/p2$
```

Step 3: run ``npm init -y`` in the terminal to create `package.json`, modify it as following:

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$ cat package.json
{
  "name": "p2",
  "version": "1.0.0",
  "description": "",
  "main": "dist/index.js",
  "scripts": {
    "start": "node dist/index.js",
    "build": "mkdir -p dist && cp -r src/* dist/"
  },
  "author": "",
  "dependencies": {
    "express": "^4.18.2"
  }
}
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$
```

Step 4: run `npm install` to install node modules.

Step 5: Build the docker image

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$ sudo docker build -t program_2 .
[+] Building 2.7s (15/15) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 425B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 28
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> [internal] load build context
=> => transferring context: 41.49kB
=> CACHED [builder 2/6] WORKDIR /app
=> CACHED [builder 3/6] COPY package.json package-lock.json ./
=> CACHED [builder 4/6] RUN npm install
=> CACHED [builder 5/6] COPY . .
=> CACHED [builder 6/6] RUN npm run build
=> CACHED [stage-1 3/6] COPY --from=builder /app/package.json ./
=> CACHED [stage-1 4/6] COPY --from=builder /app/package-lock.json ./
=> CACHED [stage-1 5/6] COPY --from=builder /app/dist ./dist
=> CACHED [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules
=> exporting to image
=> => exporting layers
=> => writing image sha256:2dbce9968101161c4e65df76b798a2b6177a304b214a563c52b42efd86ec25e0
=> => naming to docker.io/library/program_2
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$
```

Step 6: Run the docker image

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/p2$ sudo docker run -p 3000:3000 program_2  
Listening at port 3000
```

Step 7: Go to <http://localhost:3000> in the browser

